Plan & implement a smart project

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Smart Tech bootcamp

30/06/2026

Introduction and Client Requirements

You work for an IT department within Just IT. The company has purchased a new office space and has tasked you to project manage it. The new office space will need to be setup with both traditional workspace networked devices and smart IoT devices. You need to be able to install and configure the new office network meeting the needs of the client. The project needs to be completed within 4 days and requires the following:

- **Planning** – you MUST create a plan on how your project will progress. This can be done using Gantt chart and Kanban Boards.
- **Design** – you MUST use the below template to decide on the logical configuration of your network.
- **Implementation** – you MUST use the Packet Tracer to set up and configure your network. This MUST also include the IoT devices.
- **Testing** – you MUST show that your network and IoT equipment works by carrying out variety of tests.
- **Documenting** – you MUST use this document to record your activities. You MUST also create a User Guide on how to use IoT Server, Wireless DHCP Router, and how to manually and dynamically assign IP to network devices.
- **Reviewing** – you MUST review your project upon its completion.

Planning

You need to use Gantt Chart Kanban boards to plan your project. The project needs to be completed within 4 days, and you must decide on the activities that you will complete on each day. Once you have created your Gantt chart and Kanban boards, paste the screenshot evidence of each below.

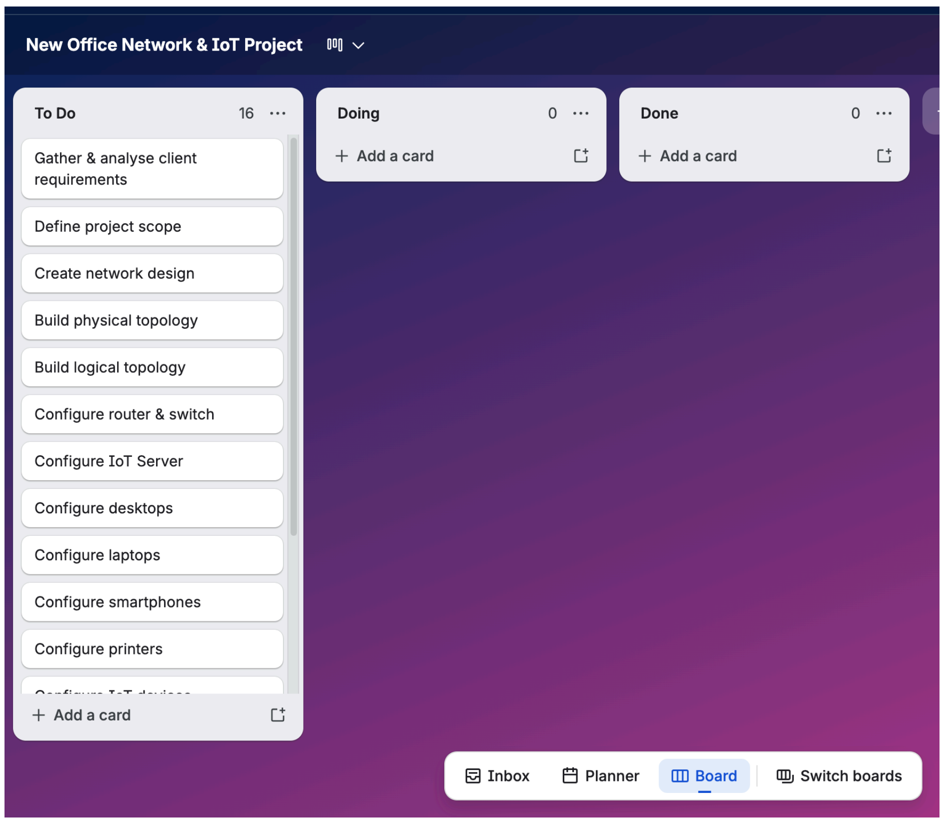
Gantt Chart

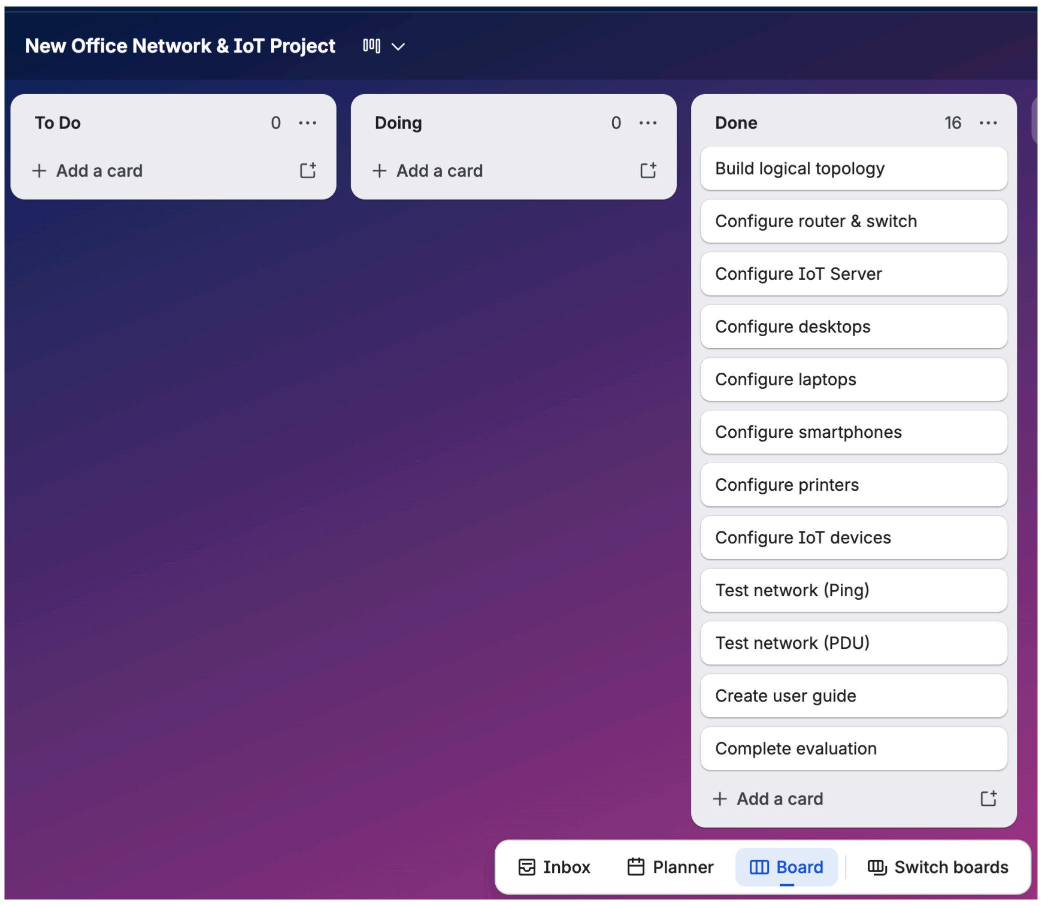
New Office Network & IoT Project GANTT CHART

PROJECT TITLE	New Office Network & IoT Project	COMPANY NAME	
PROJECT MANAGER	Nasrin Younis	DATE	

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE	PHASE ONE Day 1	Day 2	Day 3	Day 4
1	**Planning**									
1.1	Gather & analyse client requirements	Project Manager	26/05/26	26/05/26	0.5	0%				
1.2	Define project scope & objectives	Project Manager	26/05/26	26/05/26	0.5	0%				
1.3	Create Gantt chart Kanban board	Project Manager	26/05/26	26/05/26	0.5	0%				
1.4	Plan the network design	Project Manager	26/05/26	26/05/26	0.5	0%				
2	**Design**									
2.1	Create IP addressing scheme	Project Manager	27/05/26	27/05/26	0.5	0%				
2.2	Design logical network topology	Project Manager	27/05/26	27/05/26	0.5	0%				
3	**Implementation**									
3.1	Build network in Packet Tracer	Network engineer	27/05/26	28/05/26	1	0%				
3.2	Configure Router, Switches & Wireless network	Network engineer	28/05/26	28/05/26	0.5	0%				
3.3	Configure IoT server & IoT devices	Network engineer	28/05/26	28/05/26	0.5	0%				
4	**Testing & Troubleshooting**									
4.1	Test network connectivity	IT team	28/05/26	28/05/26	0.5	0%				
4.2	Test IoT devices & automation	IT team	29/05/26	29/05/26	0.5	0%				
5	**Documentation**									
5.1	Create user guide & project documentation	Project Manager	29/05/26	29/05/26	0.5	0%				
6	**Review**									
6.1	Review & evaluate project	Project Manager	29/05/26	29/05/26	0.5	0%				

Kanban board

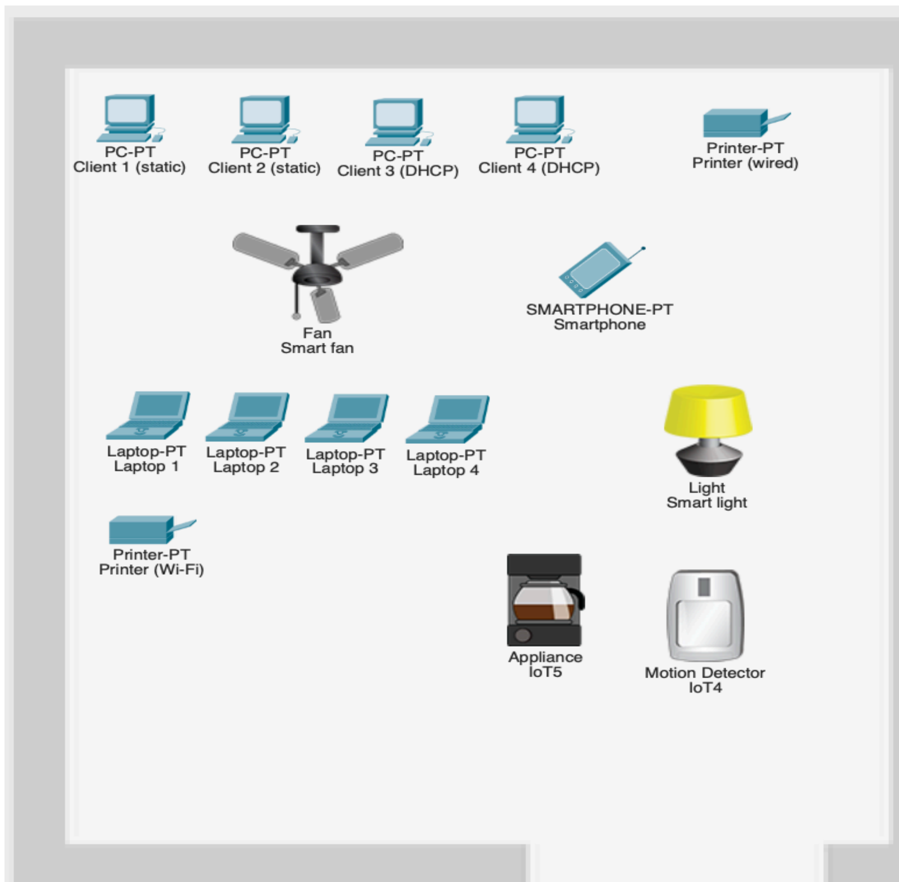




Design

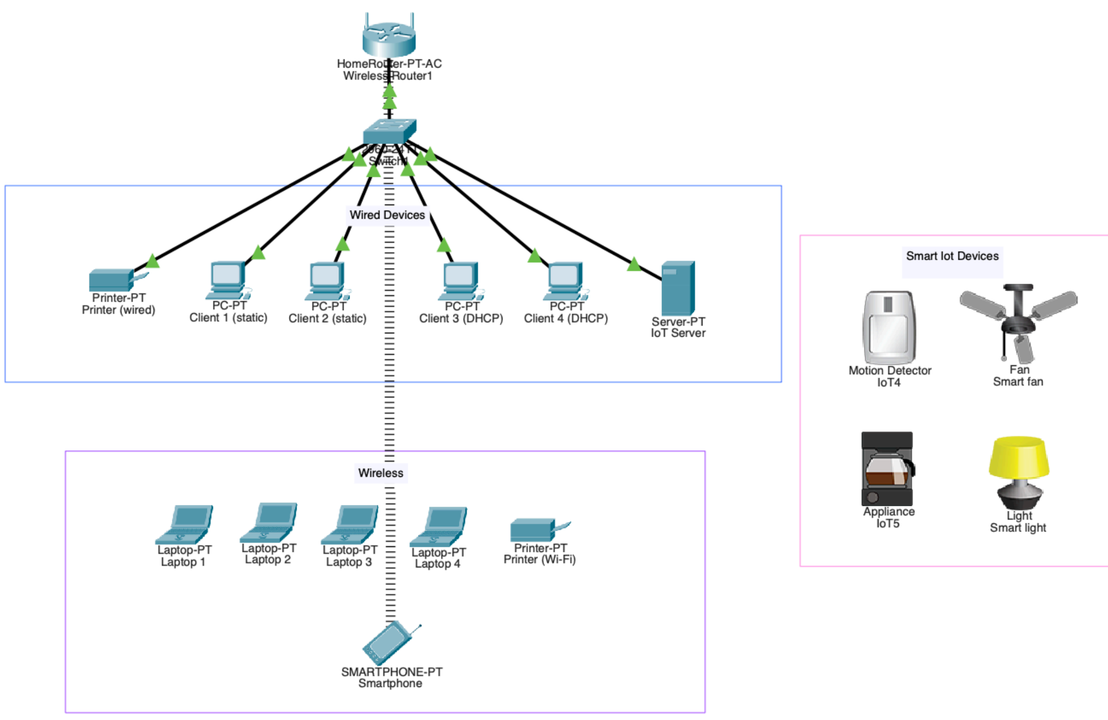
Physical topology -

Office layout

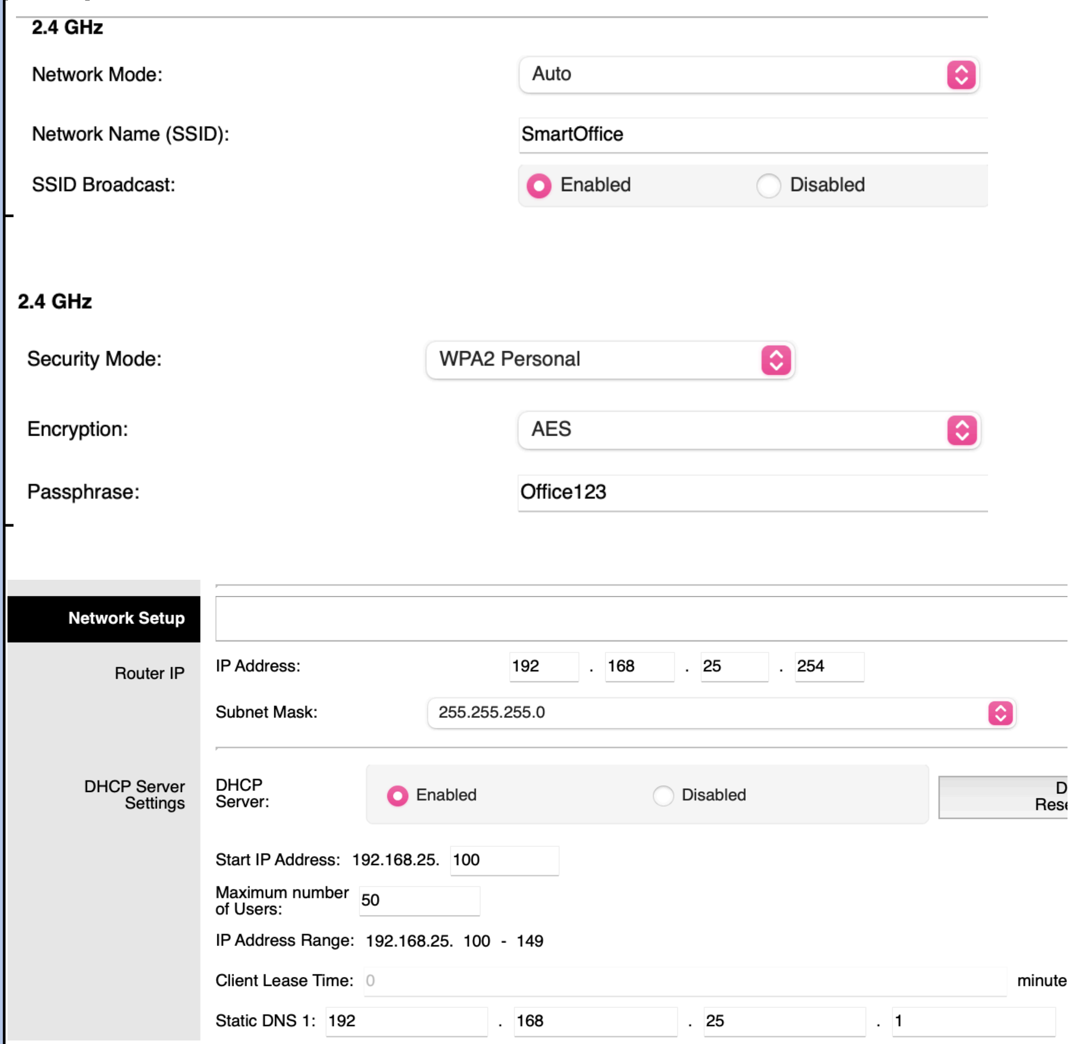


Main Wiring closet

Logical topology -

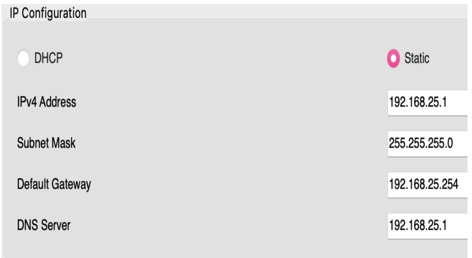


Configuration plan

Wireless Devices	**Wireless Router – Wireless Settings (SSID and Pass Phrase)**
Connection type	Wireless
	Place your snip of the wireless settings on the router including SSID and pass phrase  ***Annotation*** The wireless router has been configured with the SSID (SmartOffice) and has been secured using WPA2 personal encryption with the passphrase (Office123). This ensures that wireless connectivity is provided for laptops, smartphones, wireless printer and IoT devices.

<table>
<tr><th>Device</th><th>IoT Server</th><th>Printer 1</th><th>Printer 2</th></tr>
<tr><td>IPv4 address</td><td>192.168.25.1</td><td>192.168.25.3</td><td>192.168.25.4</td></tr>
<tr><td>Subnet Mask</td><td>255.255.255.0</td><td>255.255.255.0</td><td>255.255.255.0</td></tr>
<tr><td>DNS Server</td><td>192.168.25.1</td><td>192.168.25.1</td><td>192.168.25.1</td></tr>
<tr><td>Gateway</td><td>192.168.25.254</td><td>192.168.25.254</td><td>192.168.25.254</td></tr>
<tr><td>Connection type</td><td>Cat 5e (Ethernet)</td><td>Cat 5e (Ethernet)</td><td>Wi-Fi</td></tr>
<tr><td>Snip of IP configuration</td><td>

Place your snip here of IoT Server configuration

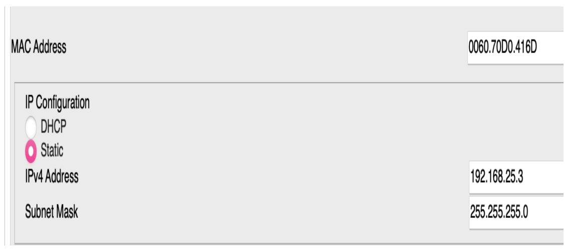


Annotation

The IoT server was configured with a static IP address (192.168.25.1), subnet mask (255.255.255.0), default gateway (192.168.25.254) and DNS server (192.168.25.1)

</td><td>

Place your snip here of printer configuration

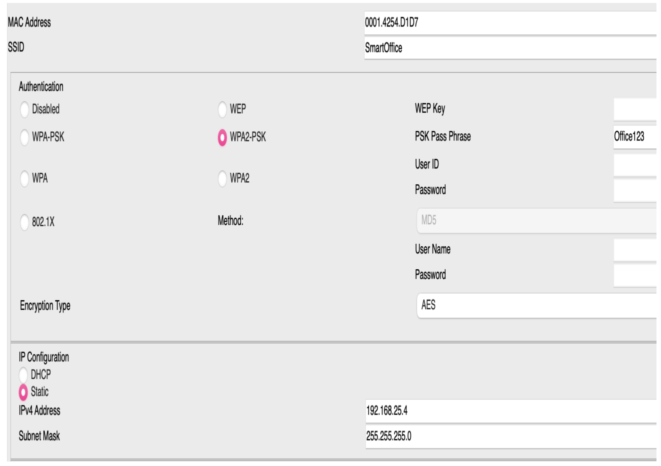


Annotation

The wired printer was configured with a static IP address of 192.168.25.3, subnet mask 255.255.255.0, and DNS server 192.168.25.1 this allows the printer to communicate reliably with other devices on the network.

</td><td>

Place your snip here of printer configuration



Annotation

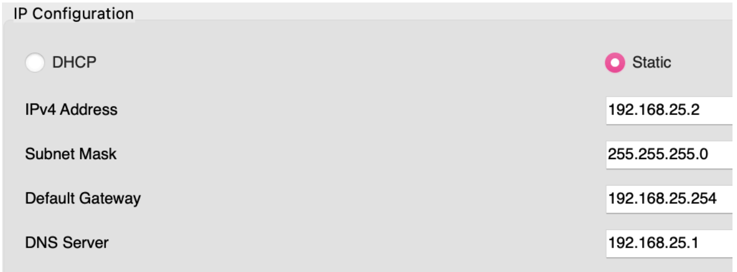
The wireless printer was configured to connect to the SmartOffice wireless network using WPA2-PSK (WPA2 Personal) security and the passphrase (Office123). A static IP address of 192.168.25.4 was also assigned with a subnet mask of 255.255.255.0, default gateway 192.168.25.254 and DNS server 192.168.25.1

</td></tr>
</table>

Device	Client 1	Client 2	Client 3	Client 4
IPv4 address	192.168.25.10	192.168.25.11	DHCP	DHCP
Subnet Mask	255.255.255.0	255.255.255.0		
DNS Server	192.168.25.1	192.168.25.1	192.168.25.1	192.168.25.1
Gateway	192.168.25.254	192.168.25.254	192.168.25.254	192.168.25.254
Connection type	Cat 5e (Ethernet)	Cat 5e (Ethernet)	Cat 5e (Ethernet)	Cat 5e (Ethernet)

Device	Client 1	Client 2	Client 3	Client 4

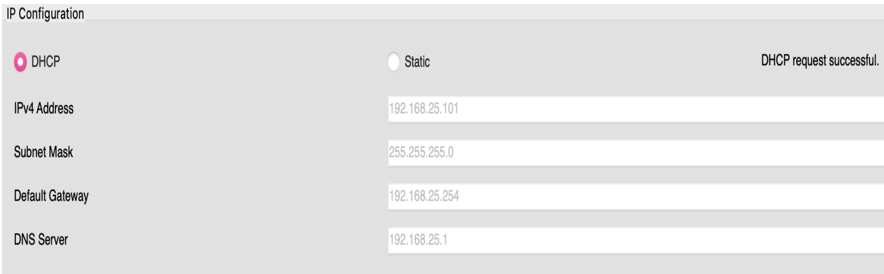
Place your snip of client 1 IP configuration



Annotation

Client 1 was configured with a static IPv4 address 192.168.25.2, subnet mask 255.255.255.0, default gateway 192.168.25.254, DNS server 192.168.25.1. A static IP address ensures the device always uses the same network address. This provides reliable communication and makes network management easier.

Place your snip of client 3 IP configuration

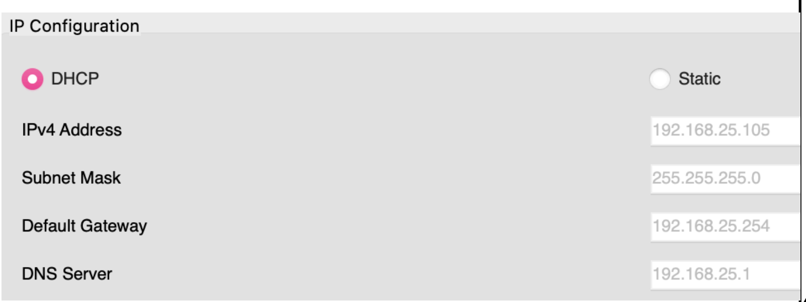


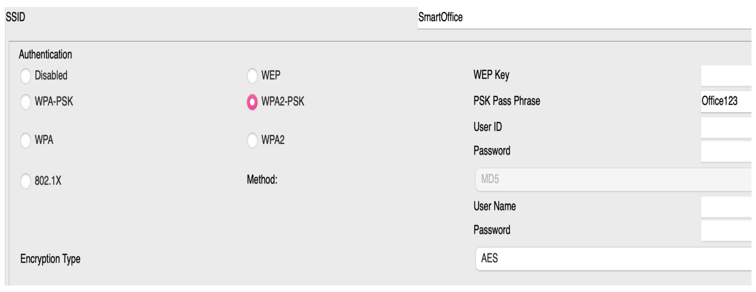
Annotation

Client 3 was configured to obtain its network settings automatically using DHCP. The device received an IP address, subnet mask, default gateway and DNS server from the wireless router's DHCP server which allows automatic network configuration and reduces the need for manual IP management.

Wireless Devices	Laptops	Smart Phones
IPv4 address	DHCP	DHCP
Subnet Mask	255.255.255.0	255.255.255.0
Connection type	Wi-Fi	Wi-Fi

Place your snip of laptop IP configuration

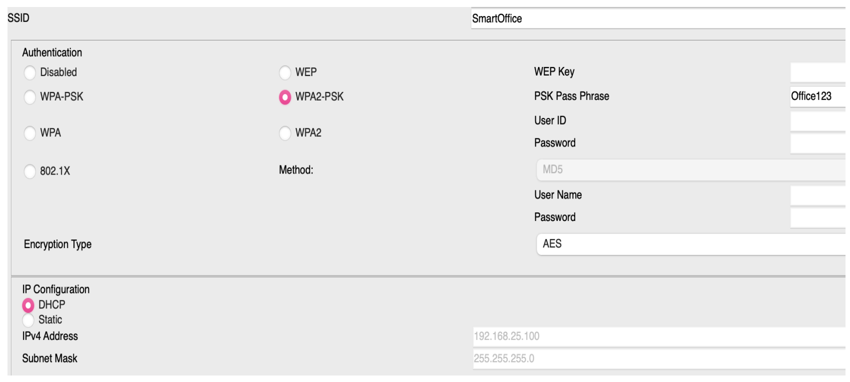




Annotation

Laptop 1 was connected to the SmartOffice wireless network using WPA2-PSK security. The device was configured to obtain its IP address, subnet mask, default gateway and DNS server automatically from the DHCP server.

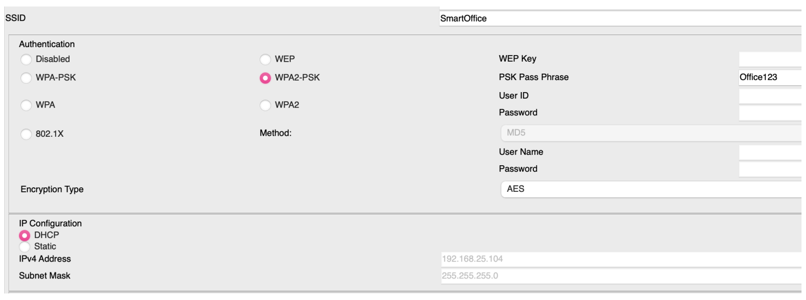
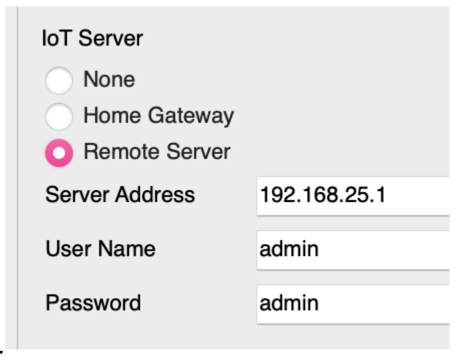
Place your snip of SMART phone IP configuration

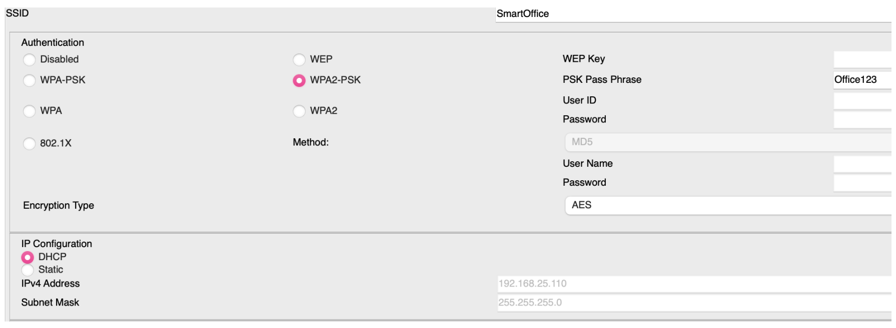
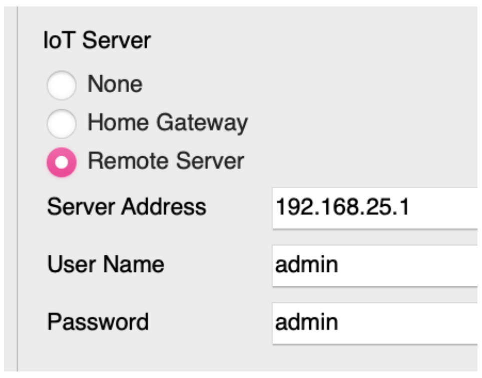


Annotation

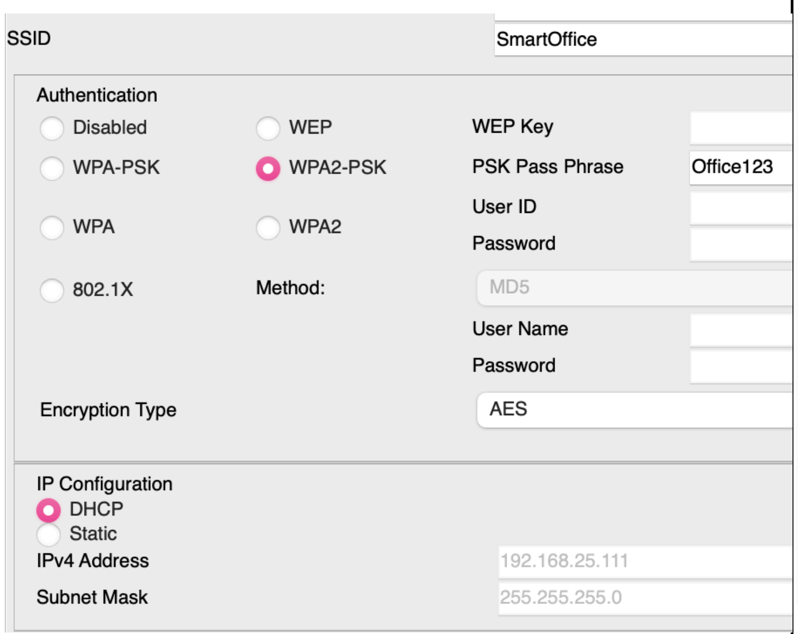
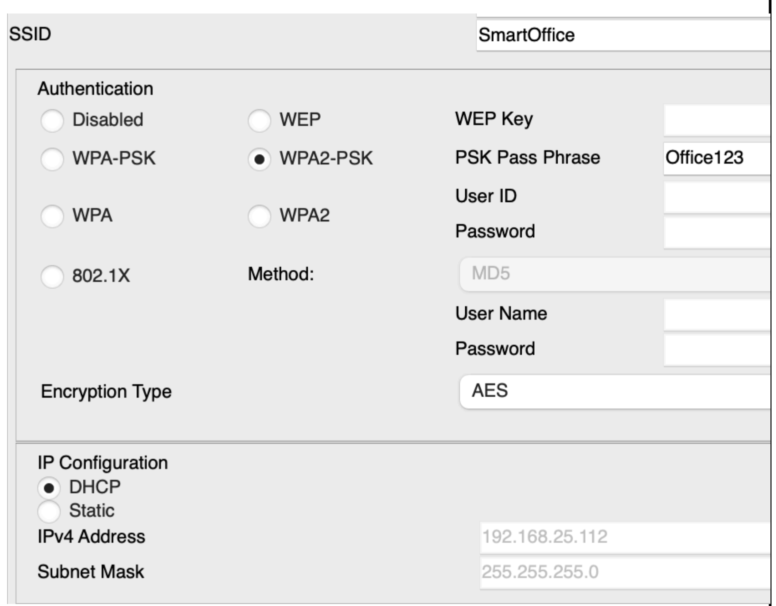
The smartphone was configured to connect wirelessly to the SmartOffice wireless network using WPA2-PSK security. DHCP automatically assigned the IP address 192.168.25.100

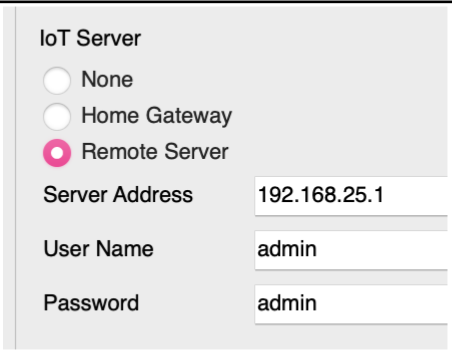
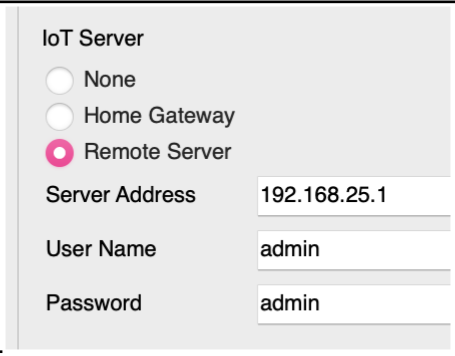
Wireless Devices	IoT Device 1 - Motion detector	IoT Device 2 - Smart fan
IPv4 address	DHCP	DHCP
Subnet Mask	255.255.255.0	255.255.255.0

<table>
<tr><td>Connection type</td><td>Wi-Fi</td><td>Wi-Fi</td></tr>
<tr><td></td><td>Place your snip of IoT configuration



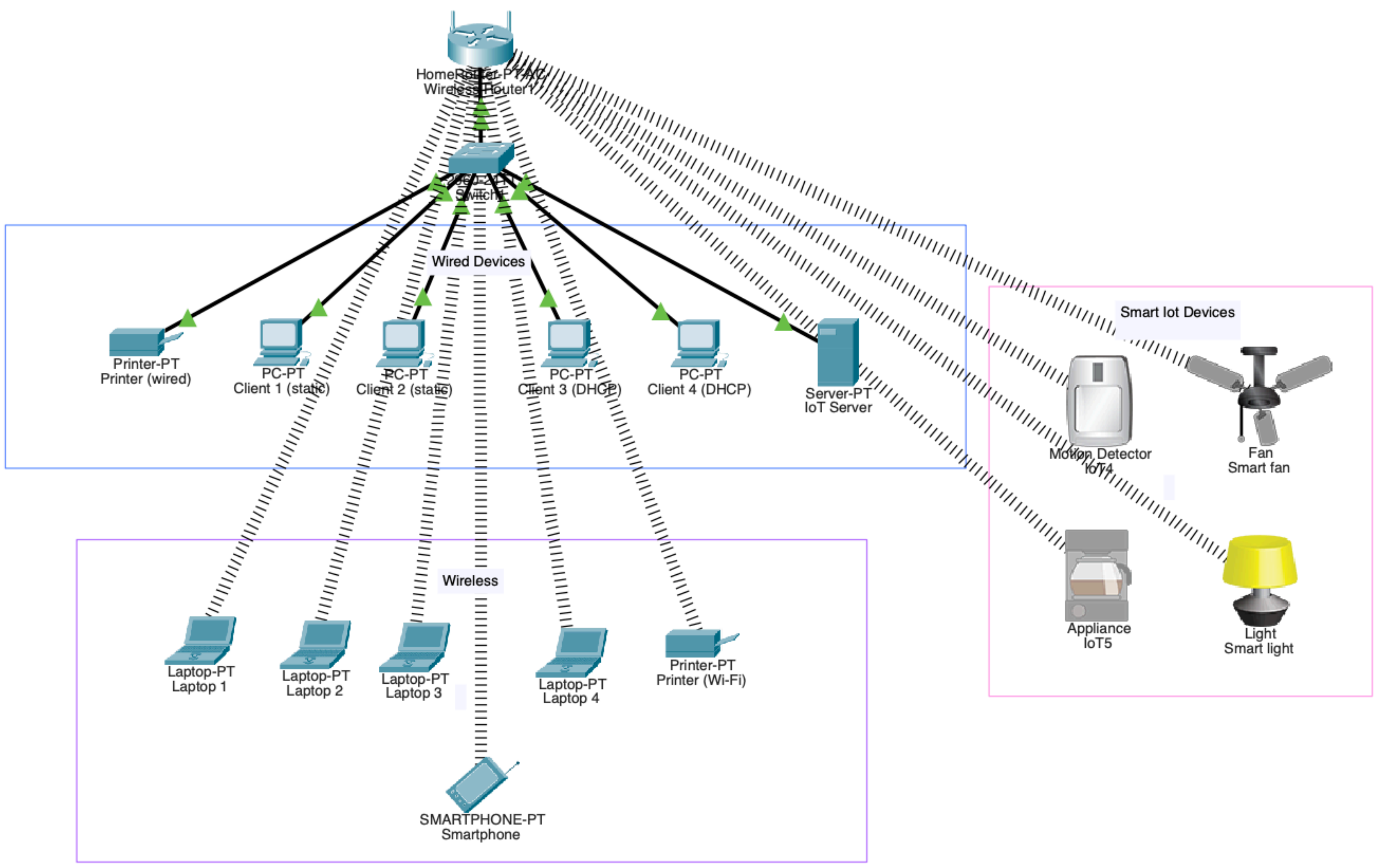
Annotation
The motion detector was connected using the wireless network WPA2-PSK security and obtained its IP address, subnet mask, default gateway and DNS server via DHCP. It was also registered with the IoT server</td>
<td>Place your snip of IoT configuration



Annotation
The smart fan was connected using the wireless network WPA2-PSK security and obtained its IP address, subnet mask, default gateway and DNS server via DHCP. It was also registered with the IoT server</td></tr>
</table>

<table>
<tr><td>Wireless Devices</td><td>IoT Device 3 - Coffee machine</td><td>IoT Device 4 - Smart light</td></tr>
<tr><td>IPv4 address</td><td>DHCP</td><td>DHCP</td></tr>
<tr><td>Subnet Mask</td><td>255.255.255.0</td><td>255.255.255.0</td></tr>
<tr><td>Connection type</td><td>Wi-Fi</td><td>Wi-Fi</td></tr>
<tr><td></td><td>Place your snip of IoT configuration

</td>
<td>Place your snip of IoT configuration

</td></tr>
</table>

	 ***Annotation*** The coffee machine was connected using the wireless network WPA2-PSK security and obtained its IP address, subnet mask, default gateway and DNS server via DHCP. It was also registered with the IoT server	 ***Annotation*** The smart light was connected using the wireless network WPA2-PSK security and obtained its IP address, subnet mask, default gateway and DNS server via DHCP. It was also registered with the IoT server

Physical topology-



Testing

PDU test 1 - PC Client 1 to IoT server

	0.002	Client 1 (static)	Switch1	ICMP
	0.002	Switch1	IoT Server	ICMP
	0.003	Switch1	IoT Server	ICMP
	0.003	IoT Server	Switch1	ICMP

Source device: PC client 1 (static IP 192.168.25.2)
Destination device: IoT server (192.168.25.1)

Result: A simple PDU (ICMP ping) was sent from PC client 1 to the IoT server. The simulation showed the packet successfully travelled through the network switch to the server and returned to the client.

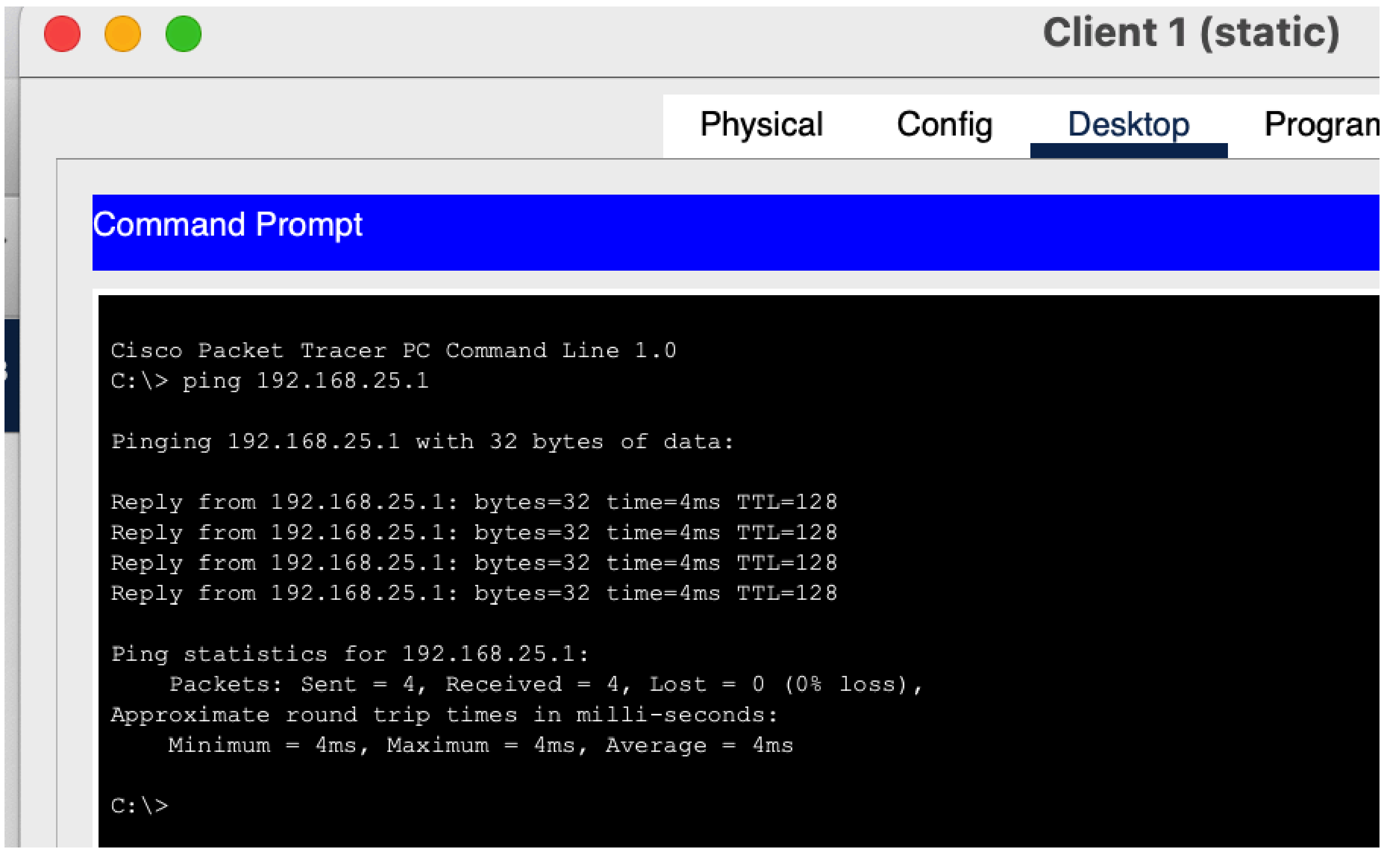
PDU test 2 - Laptop 1 to IoT server

0.008	Laptop 1	Wireless Router1	ICMP
0.009	Wireless Router1	Switch1	ICMP
0.010	Switch1	IoT Server	ICMP
0.011	IoT Server	Switch1	ICMP

Source device: Laptop 1 (DHCP)
Destination: IoT server (192.168.25.1)

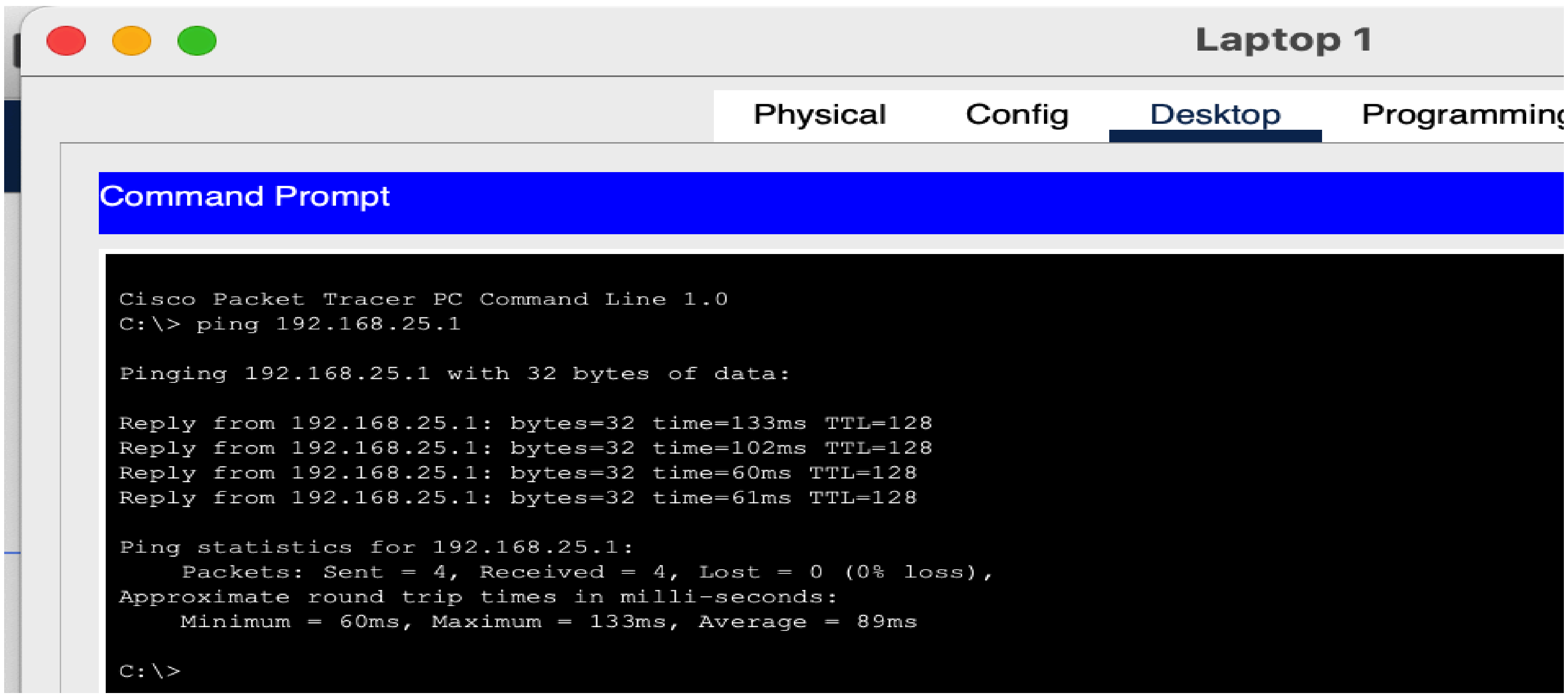
Result: A simple PDU (ICMP ping) was sent from Laptop 1, which obtained its IP address automatically via DHCP, to the IoT server. The simulation showed the packet travelling successfully through the wireless router to the network switch and then reaching the IoT server.

Ping test 1 - PC Client 1 to IoT server




```
Cisco Packet Tracer PC Command Line 1.0
C:\> ping 192.168.25.1

Pinging 192.168.25.1 with 32 bytes of data:

Reply from 192.168.25.1: bytes=32 time=4ms TTL=128
Reply from 192.168.25.1: bytes=32 time=4ms TTL=128
Reply from 192.168.25.1: bytes=32 time=4ms TTL=128
Reply from 192.168.25.1: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.25.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms

C:\>
```

Ping test 2 - Laptop 1 to IoT server



Tracert test 1 - Laptop 1 to IoT server

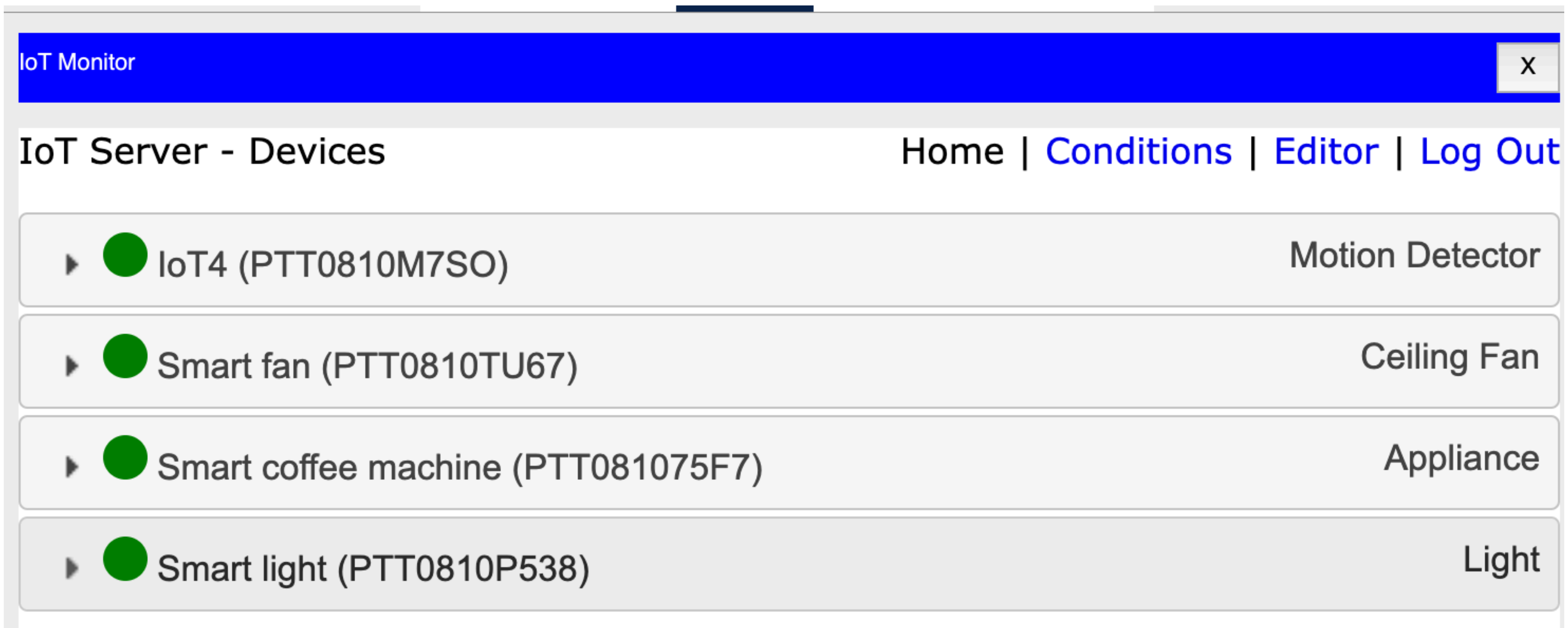
```
C:\> tracert 192.168.25.1

Tracing route to 192.168.25.1 over a maximum of 30 hops:

  1   32 ms     21 ms     37 ms     192.168.25.1

Trace complete.

C:\>
```

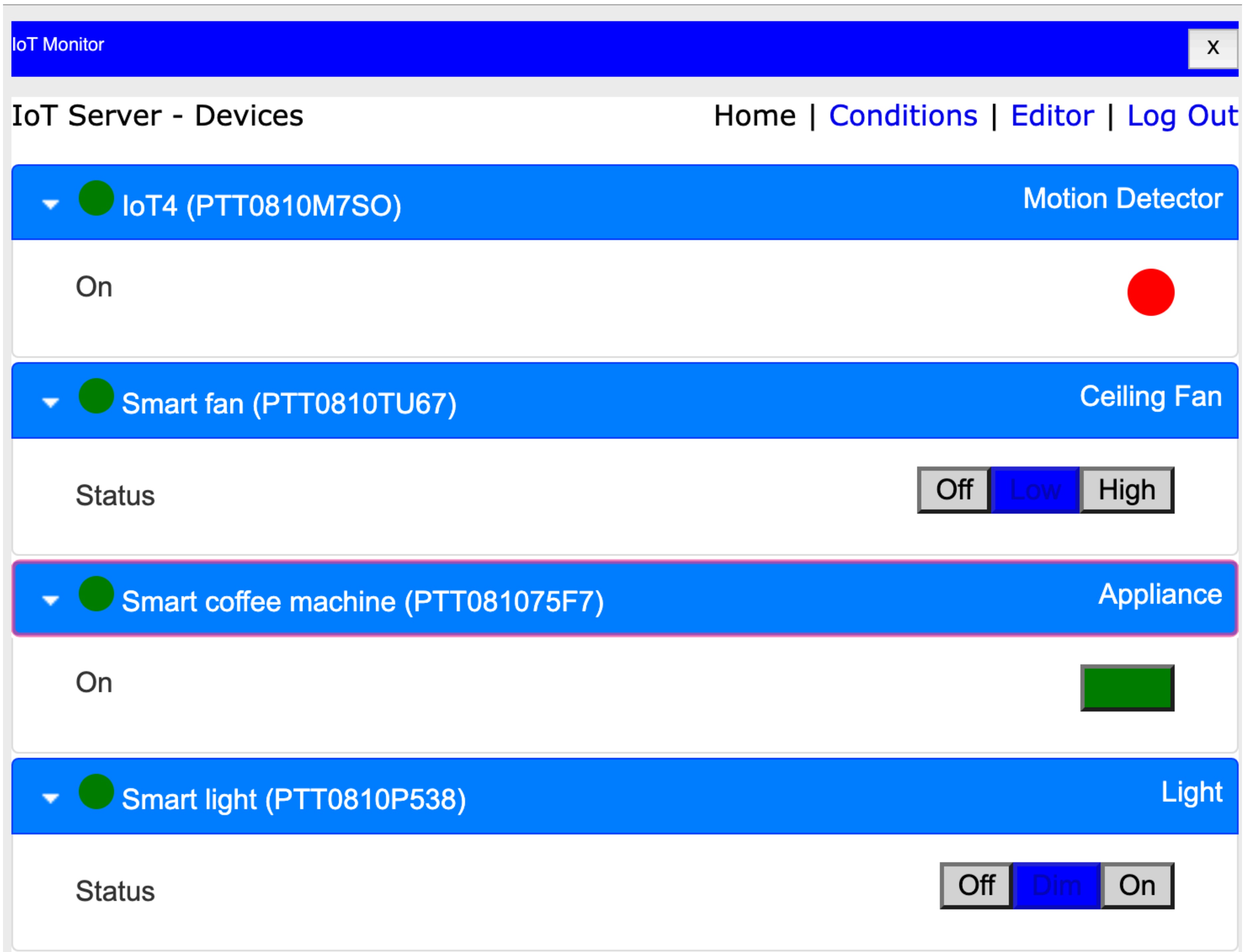
Tracert test 2 - PC Client 1 to IoT server

```
C:\> tracert 192.168.25.1

Tracing route to 192.168.25.1 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.25.1

Trace complete.

C:\>
```

Monitor & Control

IoT monitor app -





Post Project Evaluation

I completed setting up Smart Office Network and IoT. Used Cisco Packet Tracer with wired and wireless end devices. I checked IPs, static and DHCP, to make sure they worked. All the devices can communicate with each other.

I tested connectivity with Ping, Tracert, and PDU simulation. All tests showed devices talking across the network.

The smart lights, fan, coffee machine, and motion detector all connected to the Wi-Fi. I managed them through the IoT Management App.

Issues I encountered:

The wireless printer wouldn't connect at first. Turns out the wireless module wasn't set up right.
Some IoT devices didn't show up in the app until we registered them with the IoT Server. I had to go back and fix that.
I had to plan static IPs carefully so they wouldn't conflict with DHCP.
Early PDU simulations sent out ARP traffic before the pings. The network just needed time to figure out the MAC addresses first.

What went well:

Got the wired and wireless devices set up fine.
Static and DHCP addresses worked like they should.
Secured the Wi-Fi using WPA2-PSK.
Printers were accessible, which is key.
The app lets us manage and watch the IoT devices.
Ping, Tracert, and PDU tests confirmed the network was fine.

If we were doing this for real, some things to do better next time:

Use VLANs to split up departments makes it safer.
Set up firewalls for extra security.
Keep backup switches or routers in case one goes down. This ensures everything is running smoothly and is reliable.
Back up the IoT Server automatically.
We can add more smart IoT devices, such as cameras, locks, or smart doors etc.

Overall, the objective was achieved and all the requirements have been met.The smart office network was secure and worked. It showed how you can get reliable connections, manage devices easily, and add some automation with wired, wireless, and IoT devices.